The Hamiltonian of the toy model is:

$$\hat{H} = -J_x \sum_{\langle ij \rangle_x} \hat{S}_i \cdot \hat{S}_j - J_y \sum_{\langle ij \rangle_y} \hat{S}_i \cdot \hat{S}_j - g\mu_B B \sum_i \hat{S}_i^x \tag{1}$$

Set $h = \mu_B B$. Heisenberg EOM is $\frac{d}{dt}\hat{S}_i = \frac{1}{i\hbar}[\hat{S}_i, \hat{H}]$.

$$\frac{d}{dt}\hat{S}_i = \hat{S}_i \times \left(J_x(\hat{S}_{i+\mathbf{e}_x} + \hat{S}_{i-\mathbf{e}_x}) + J_y(\hat{S}_{i+\mathbf{e}_y} + \hat{S}_{i-\mathbf{e}_y}) + h\mathbf{e}_x \right) \tag{2}$$

Take classical limit $\mathbf{S} = S\mathbf{e}_x + s^y\mathbf{e}_y + s^z\mathbf{e}_z$:

$$\dot{s}_i^y = s_i^z h - S(J_x(s_{i+\mathbf{e}_x}^z + s_{i-\mathbf{e}_x}^z) + J_y(s_{i+\mathbf{e}_y}^z + s_{i-\mathbf{e}_y}^z)) \tag{3}$$

$$\dot{s}_i^z = -s_i^y h + S(J_x(s_{i+\mathbf{e}_x}^y + s_{i-\mathbf{e}_x}^y) + J_y(s_{i+\mathbf{e}_y}^y + s_{i-\mathbf{e}_y}^y)) \tag{4}$$

Take continue limit (Landau-Lifshitz equation, but we will not use this equation):

$$\frac{d}{dt}M(x,y) = M(x,y) \times \left(J_x a^2 \partial_x^2 M(x,y) + J_y a^2 \partial_y^2 M(x,y) + h\mathbf{e}_x \right) \tag{5}$$

We use plane wave method to solve the classical equation:

$$s^y(r) = \exp(ik \cdot r) \sum_G s_G^y \exp(iG \cdot r)$$

$$s^z(r) = \exp(ik \cdot r) \sum_G s_G^z \exp(iG \cdot r)$$

and then we have:

$$-i\omega s_G^y = s_G^z[h - 2S(J_x \cos(k_x a) + J_y \cos(k_y a))] \tag{6}$$

$$-i\omega s_G^z = -s_G^y[h - 2S(J_x \cos(k_x a) + J_y \cos(k_y a))] \tag{7}$$

Set $\tilde{s}_G = s_G^y \pm i s_G^z$, then we have $\omega = \mp[2S(J_x \cos(k_x a) + J_y \cos(k_y a)) - h]$. We choose $k_x = \frac{\pi}{2}, k_y = \frac{\pi}{2}$ and use the gauge $\tilde{s}_G = s_G^y \pm i s_G^z$, the dispersion relation becomes:

$$\omega = (h - \pi S J_x - \pi S J_y) + 2SJ_x k_x + 2SJ_y k_y + \omega_{nonlin} \tag{8}$$

$$\omega_{nonlin} = -\frac{1}{3}S[J_x(k_x - \frac{\pi}{2})^3 + J_y(k_y - \frac{\pi}{2})^3] \tag{9}$$

So for the wave packet surrounding $(\frac{\pi}{2}, \frac{\pi}{2})$, the solution of EOM can be written as:

$$\begin{aligned}\tilde{s}(x,y,t) &= \int\int dk_x dk_y \psi(k_x,k_y) e^{ik_x x + ik_y y} e^{-i2SJ_x k_x t - i2SJ_y k_y t - i\omega_{nonlin} t} \\ &= \int\int dk_x dk_y \psi(k_x,k_y) e^{ik_x(x-2SJ_x t) + ik_y(y-2SJ_y t)} e^{-i\omega_{nonlin} t}\end{aligned}$$

Therefore we can introduce a compensation phase shift to mitigate the dispersion caused by the nonlinear term: $e^{i\omega_{nonlin} T}$, where T is a positive constant.